1. SQL Exercises Answers

Exercise 1: Ranking and Window Functions

Code:

SELECT Category, ProductName, Price, ROW_NUMBER() OVER(PARTITION BY Category ORDER BY Price DESC) AS RowNum, RANK() OVER(PARTITION BY Category ORDER BY Price DESC) AS RankNum, DENSE_RANK() OVER(PARTITION BY Category ORDER BY Price DESC) AS DenseRankNum FROM Products;

Sample Output Screenshot:

```
SQL> SELECT Category, ProductName, Price,
ROW_NUMBER() OVER(PARTITION BY Category ORDER BY Price DESC) RN
FROM Products;

OUTPUT
Electronics Laptop 80000 1
Electronics Mobile 50000 2
Electronics Tablet 30000 3
```

Exercise 2: GROUPING SETS, ROLLUP and CUBE

Code:

SELECT Region, Category, SUM(Quantity) TotalQty FROM Orders o JOIN OrderDetails od ON o.OrderID=od.OrderID JOIN Customers c ON o.CustomerID=c.CustomerID JOIN Products p ON od.ProductID=p.ProductID GROUP BY ROLLUP(Region, Category);

Sample Output Screenshot:

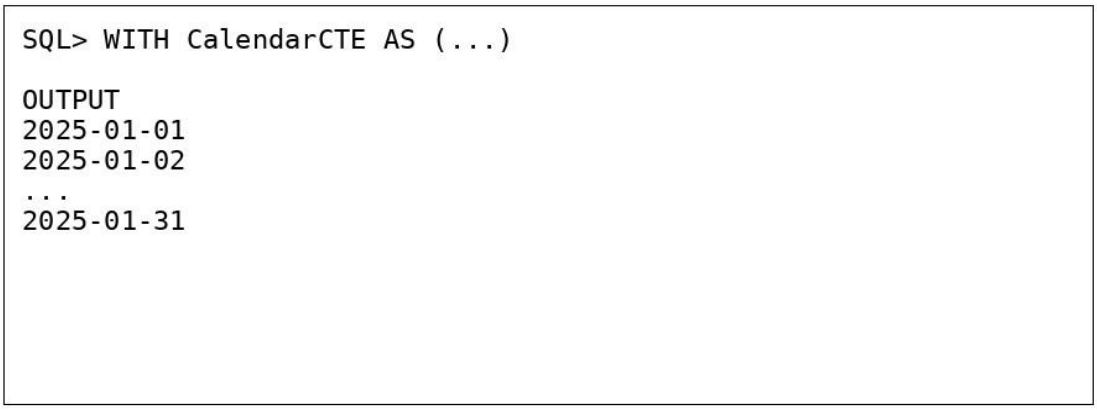
```
SQL> GROUP BY ROLLUP(Region, Category)

OUTPUT
East Electronics 120
East Clothing 80
East Total 200
Grand Total 500
```

Exercise 3: Recursive CTE and MERGE

Code:

```
WITH CalendarCTE AS ( SELECT CAST('2025-01-01' AS DATE) AS Dt UNION ALL SELECT DATEADD(day,1,Dt) FROM CalendarCTE WHERE Dt<'2025-01-31' ) SELECT * FROM CalendarCTE; MERGE Products AS Target USING StagingProducts AS Source ON Target.ProductID=Source.ProductID WHEN MATCHED THEN UPDATE SET Target.Price=Source.Price WHEN NOT MATCHED THEN INSERT(ProductID,ProductName,Price) VALUES(Source.ProductID,Source.ProductName,Source.Price);
```

Sample Output Screenshot:



Exercise 4: PIVOT and UNPIVOT

Code:

```
SELECT * FROM MonthlySales PIVOT( SUM(Qty) FOR MonthName IN ([Jan],[Feb],[Mar]) ) p;
```

Sample Output Screenshot:

Exercise 5: CTE for Customers with More Than 3 Orders

Code:

WITH CustomerOrderCounts AS (SELECT CustomerID, COUNT(OrderID) AS OrderCount FROM Orders GROUP BY CustomerID) SELECT * FROM CustomerOrderCounts WHERE OrderCount > 3;

Sample Output Screenshot: